



HDD and SSD repair in India

ReStor opened India's first HDD and SSD repair/refurbishment facility in Gurgaon, Delhi-NCR
<http://dgit.in/150yLf5>

Memory cards demystified

We all use memory cards but do we understand them? We demystify these little plastic slabs for you <http://dgit.in/16j0cRv>

Portable HDDs Test

HOW WE TESTED

For performance testing, we divided our tests into two categories – synthetic tests and real world tests. Synthetic benchmarks give an overview of the raw performance for every drive. Our tests included a full run of PC Mark 8 storage benchmark which gave us thorough results along with the time taken to run application based tests using Adobe Photoshop, Adobe After Effects, Microsoft Powerpoint and Battlefield 3 to name a few. Next we used Crystal Disk Mark to run 3 passes of 0x00 Fill and Random Fill. We took into account the sequential and 512K read and write scores using a test sample of 100MB. The next synthetic benchmark used was the SiSoft Sandra 2013 to obtain drive index read and write scores along with the access time for data read and write. Real world tests give an accurate account of what a user should expect from a particular drive. So for real world scores, single file read and write tests were done using a 4GB and a 20GB file separately. For multiple file transfer tests, we used similar data sets totalling up to a collection of 4 GB and 20 GB each, of assorted files. Intra drive tests were also done to know about transfer rate for read and write within the

same drive by creating two partitions. We used a digital stopwatch to record the file write and file read timings for all the single and assorted read/write tests.

The test rig that we used for the comparison test had the following configuration:

- Processor: Intel 4th Gen Core i7 4770K;
- Motherboard: ASRock Z87 Extreme 6;
- RAM: ADATA AX3U2133XW4G10-2X (4 DIMMs x 4GB=16GB total)
- Primary storage: 2 x Kingston Hyper-X 120GB SSD SH100S3-120G in RAID;
- GPU: Onboard;
- CPU Cooler: Cooler Master Hyper 212X;
- Power Supply: Cooler Master Silent Pro 850;
- Cabinet: Cooler Master HAF-XM;
- Monitor: DELL S2740L;
- Operating System: Windows 7 SP1.

We used top of the line components to prevent bottlenecks in performance to ensure the tests go smoothly and the results are accurate. Once we were done, we set parameters to judge for aspects such as Build Quality, Performance and features for comparison. We assessed the build quality of each drive by taking into account their body material, padding and feet of the

drive. So a drive that had a full metal body with good finishing received more points than the drives having plastic bodies. If a drive was padded with a rubber outer covering then it received extra points for that as well. Some grip is always good for portable HDDs especially when they are kept on uneven surfaces or table tops and the feet of the drive help in that sense, so we assigned points to drives that had rubber or silicone feet.

For comparing drives according to their features, we assigned points to each drive based on a few areas. If a drive supports or comes with backup, encryption and social media backup utilities then it got points for that. Everything is moving towards cloud storage so the drives that allow you to store or access content from various cloud storage services fared well in terms of points for features. Performance is something that defines a product, so we gave performance the maximum weightage in the comparison. Finally we categorically divided the drives according to their size. Two categories were made, one with 500GB drives and the other with drives having a storage space of 1TB or more.

the drives to complete small file transfers with great ease. Compared to previous generation drives, the performance increase is enormous and the 5Gbit/s USB 3.0 threshold was easy to hit.

Design wise, the ADATA HE720 was the most sophisticated looking. It is completely enclosed in a full metal stainless steel body and is one of the slimmest drives we've come across. The drive has a brushed metal finish coupled with a glossy metal band around the middle region. It has a one-touch backup button at the front that triggers the ADATASync utility which comes pre-installed on the drive. This drive also supports ADATA's HDD-to-Go utility that can be used to install and run portable software directly from the HE720.

The Buffalo Ministation Thunderbolt HD-PATU3 was

another well built drive and was the only one to sport a USB 3.0 port and Apple's Thunderbolt interface. The drive is encased inside an aluminium casing which covers the sides and the bottom. The top part of the drive is made out of translucent plastic and is fixed well. Though bigger and bulkier than all other 500GB drives in the test the Buffalo Ministation is built sturdily. And has a design reminiscent to that of apple products. It comes with a single Thunderbolt port which means that you can only daisy-chain this drive with other Thunderbolt compatible products. They've bundled a Thunderbolt cable with the drive as well. Thunderbolt as a standard offers a bandwidth of up to 10Gbit/s but in order to compare its results with the other drives in the test we used the USB 3.0 interface only. The

HD-PATU3 has support for Mac Time machine backup software.

The SimpleSave P2050B from HP also had a brushed aluminium finish but only on some sides of the metal casing. The drive is designed to mimic the shape of a book with one side made of metal and the other in plastic. However, we would've liked a full metal body. The HP 2050B's PC Mark 8 scores were lower than most other 500GB drives and fared

well in a few real world tests. But synthetic tests didn't put the drive much ahead of the competition.

Seagate's Slim 500GB drive has a standard rectangular shape made out of plain black thin metal. The drive comes with a SATA-to-USB 3.0 adapter attached to it. The main idea behind providing a removable adapter with the drive is that you get the flexibility to remove the included adapter and purchase a Thunderbolt or FireWire based adapter and use this drive with any of the three interfaces. Seagate Dashboard software is easy to navigate through and allows you to perform instant and scheduled backup operations. Performance wise the Seagate Slim STCD500400 was the least performing of the pack but its features helped redeem itself. Given a low street price of just ₹4,150 this drive is worthy of consideration.



ADATA DashDrive HE720